tech-stack | Status: current | App v1.0.20 | Reviewed 2026-10-24

Technical Stack Overview

Versions come from `Android/src/gradle/libs.versions.toml` unless noted. When you upgrade a library, update this table.

1. Platform and toolchain

Item	Value	Source
Language	Kotlin 2.2.21 (`kotlin.code.style=official`)	toml
JVM target	Java 11 (`sourceCompatibility` / `targetCompatibility`, `jvmTarget`)	`app/build.gradle.kts`
JDK used by CI	Temurin 21	`build_android.yaml`
Gradle	9.2.1 (wrapper)	`gradle-wrapper.properties`
Android Gradle Plugin	9.0.1	toml
KSP	2.3.6	toml
compileSdk / targetSdk / minSdk	37 / 37 / 31 (Android 12)	`app/build.gradle.kts`
Modules	`:app` only	`settings.gradle.kts`
Repositories	`google()`, `mavenCentral()` (`FAIL_ON_PROJECT_REPOS`)	`settings.gradle.kts`
Gradle JVM heap	`-Xmx2048m`	`gradle.properties`

2. Gradle plugins

Plugin	Purpose
`com.android.application`	App build
`org.jetbrains.kotlin.plugin.compose`	Compose compiler
`org.jetbrains.kotlin.plugin.serialization` (2.0.21)	kotlinx.serialization
`com.google.protobuf` (0.10.0, protoc 4.26.1, `java` + `kotlin` lite)	Proto codegen
`com.google.dagger.hilt.android` (2.60.1)	DI
`com.google.devtools.ksp`	Hilt and Moshi codegen
`com.google.android.gms.oss-licenses-plugin` (0.11.0)	Licenses screen

Plugin	Purpose
`com.google.gms.google-services` (4.4.3)	**`apply false`**. Needs `google-services.json`, which is not in the repo.

3. Libraries by concern

Concern	Library	Version
UI	Jetpack Compose BOM	2026.02.00
	Material 3, Material Icons Extended	BOM / 1.7.8
	Navigation Compose	2.8.9
	Activity Compose	1.10.1
	Core SplashScreen	1.2.0-beta01
	compose-richtext (Markdown, Material3 + CommonMark)	1.0.0-alpha02
	Coil Compose	2.7.0
Lifecycle	lifecycle-runtime-ktx, lifecycle-process	2.8.7
DI	Hilt Android, Hilt Navigation Compose	2.60.1 / 1.3.0
LLM runtime	LiteRT-LM Android (`com.google.ai.edge.litertlm:litertlm-android`)	0.18.0
	ML Kit GenAI Prompt (AICore)	1.0.0-beta2
Classical ML	Play Services TFLite Java / GPU / Support	16.4.0
	MediaPipe Tasks Vision (InteractiveSegmenter)	1.0.0
	MediaPipe Tasks Retrieval (UniversalEmbedder, SemanticRetriever, SqliteVectorStore)	1.1.0
Agent / MCP	MCP Kotlin SDK (`io.modelcontextprotocol:kotlin-sdk`)	0.8.0
	Ktor Client Core + Android (SSE)	3.4.3
Persistence	DataStore (proto)	1.1.7
	Protobuf Java/Kotlin Lite	4.26.1
Background	WorkManager (`work-runtime-ktx`)	2.12.0
Serialization	Gson	2.14.0
	Moshi Kotlin + codegen	1.15.2
	kotlinx-serialization-json	1.7.3
	kotlin-reflect, kotlin-metadata-jvm (forced)	2.4.0
Media	CameraX core/camera2/lifecycle/view	1.4.2

Concern	Library	Version
	Media3 ExoPlayer / Transformer / UI	1.11.1
	ExifInterface	1.4.1
Web	AndroidX WebKit (`WebViewAssetLoader`)	1.14.0
Auth	AppAuth (`net.openid:appauth`)	0.11.1
Security	AndroidX Security Crypto	1.1.0 (**declared, unused**)
Firebase	BoM, Analytics, Messaging	33.16.0
Licenses	play-services-oss-licenses	17.5.1

Test dependencies are declared but there are **no test sources** (see 06-testing): JUnit 4.13.2, AndroidX JUnit 1.2.1, Espresso 3.6.1, Compose `ui-test-junit4` , `hilt-android-testing` .

4. Data and assets formats

Format	Where
`.litertlm`	LiteRT-LM model bundles: all LLMs and EmbeddingGemma-2 in the 1.0.20 allowlist
`.task`	MediaPipe task bundles: the Scrapbook segmenter (`interactive_segmentation.task`), legacy LLMs, and importable models
`.zip`	Optional extra data (e.g. EmbeddingGemma-2 `samples.zip` sample album), unzipped after download
Protobuf (`*.proto`)	DataStore schemas, Hugging Face model cache, metrics
JSON	Model allowlists, skill allowlist, JS skill results, intent parameters
`SKILL.md` + HTML/JS	Agent Skills

5. External services

Service	Use	Required?
GitHub raw (`raw.githubusercontent.com`)	Model allowlist	Yes for the first catalog load; cached afterwards
GitHub API (`api.github.com`)	Latest-release check	No
GitHub Pages	Hosting `skills/` (from the `static.yml` workflow)	Only for remote/featured skills

Service	Use	Required?
Hugging Face (`huggingface.co`)	Model files, model search API, OAuth	Yes for downloads
Google Play services / AICore	AICore models, TFLite	Only on supported devices
Firebase Analytics / FCM	Telemetry, push	No (absent in OSS builds)
User-configured MCP servers and skill URLs	Agent tools	No

6. Native and hardware acceleration

- LiteRT-LM backends: CPU, GPU (OpenCL), NPU/TPU (vendor libraries; per-SoC model files via `socToModelFiles`).
- The manifest declares `android.hardware.npu` as optional, plus optional native libraries `libOpenCL.so`, `libcdsprpc.so` (Qualcomm DSP), `libedgetpu_*.so` (Google Tensor), `libvndksupport.so`.
- The MediaPipe embedder supports NPU, GPU, and CPU delegates.